

PROJECT TITLE

Tamra

A project submitted for the Data Analysis 1 course (DS2313)

COURSE INSTRUCTOR

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ABSTRACT

In this project, we built an AI model to classify 13 different types of dates. We used a dataset of 2,613 images (1,829 for training and 784 for testing). We used some Deep Learning models like EfficientNetB3 and EVA02. We applied techniques like Data Augmentation and Fine-Tuning to improve accuracy. Our final model achieved a Macro F1-Score of 0.838, ranking us #8 on the competition leaderboard.

INTRODUCTION

The goal of this project was to automatically identify Saudi date varieties using computer vision. This is important for the agricultural industry specifically in our country to help sort and label dates correctly. We participated in the SODA Challenge hosted by Umm Al-Qura University on Kaggle. Our main research question was: "How can we use Machine Learning to accurately classify date images?"

METHODOLGY

1. Data Source & Description

- Source: SODA Challenge hosted on Kaggle by UQU College of Computing.
- Content: The dataset has 13 classes of dates.
- Size: 2,613 total images (1,829 training, 784 testing).
- Balance: Each class is balanced with about 140–141 images.
- Format: Images are in JPG and PNG formats.

2. Data Preprocessing

Cleaning: We checked for corrupt files and found none, because it is already cleaned and ready to use.

Steps:

1. Converted labels to strings for the model.
2. Fixed image extensions (some were .jpg, some were .png).
3. Used Class Weights to make sure the model treats all 13 types of dates equally.
4. Split the data into 85% for training and 15% for validation.

3. Algorithm/Model Selection

- Models used: We started with MobileNetV2 as a baseline. Then we moved to EfficientNetB3 and finally EVA02.
- Why: These models are "Pre-trained" on millions of images, which helps them learn faster and better for our date images.

4. Training Procedure

- Step 1 (Transfer Learning): We froze the main part of the model and trained only the last layer for 20 epochs.
- Step 2 (Fine-Tuning): We unfroze all layers and trained with a very low learning rate (0.00001) for 15 epochs to get the best accuracy.
- Optimization: We used the Adam optimizer and saved the best model using "ModelCheckpoint."

5. Evaluation Strategy

- Metric: We used the Macro F1-Score because it is the best way to measure performance on balanced datasets.
- Robustness: We used K-Fold Cross Validation to make sure our results were reliable and not just lucky.
- TTA: We used Test Time Augmentation (TTA), which means we showed the model slightly different versions of the same test image and averaged the answers to get a better result.

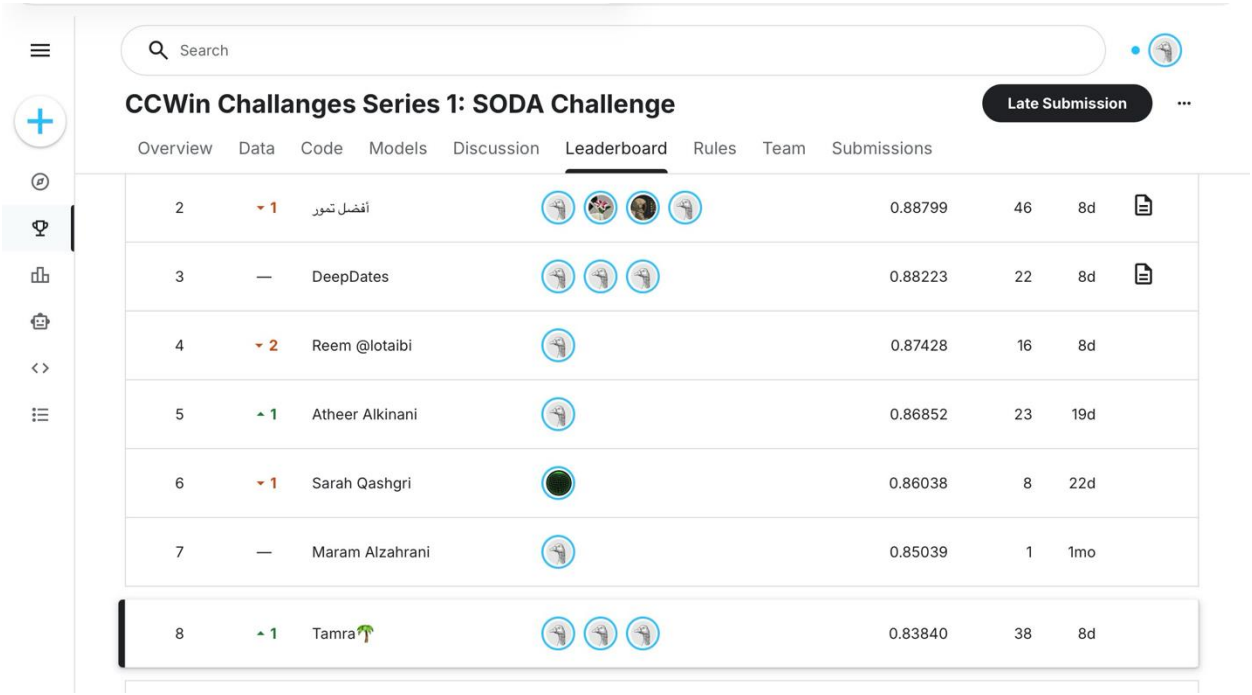
RESULTS AND DISCUSSION

Our models showed great improvement over time. The baseline MobileNetV2 only got a score of 0.411. By switching to EfficientNetB3 and using Fine-Tuning, we reached 0.789. Our best result came from the EVA02 model with MixUp/CutMix, which reached 0.838.

Model	**Technique**	**F1-Score**
MobileNetV2	Baseline	0.411
EfficientNetB3	Transfer Learning	0.692
EfficientNetB3 + TTA	5x TTA	0.786
EfficientNetB3 + FT	Fine-Tuning	0.789
EVA02 + MixUp	Final Best	0.838

The results show that Data Augmentation (like rotating and flipping images) was very important. Without it, the model would overfit. Also, TTA gave us a small but consistent boost in our final ranking.

Figure 1: Team Ranking and Final Score on the SODA Challenge Leaderboard



CONCLUSION

In conclusion, we successfully built a high-performing model for the SODA Challenge. We reached rank #8 with an F1-Score of 0.838 . We learned that Transfer Learning and Fine-Tuning are very powerful for image classification. For future work, we would use even larger models (like EfficientNetV2L) or combining multiple models together (Ensembling) to reach the #1 place.

Finally, we aspire to participate in more challenges and competitions that represent and benefit our country and help us learn more.

REFERENCES

- Data Source: UQU College of Computing (2026). SODA Challenge: Saudi Organic Dates Classification. Kaggle.
- AI agents for suggestions and improvements.
- Machine Learning Course Notes, Department of Data Science, Umm Al-Qura University, 2026

THE EXPERIENCES AND SKILLS ACQUIRED BY THE TEAM MEMBERS

Student Name	**Experiences**	**Skills**
Nouf bin Taleb	**Using Kaggle website and learning about different models and how they perform differently.**	**Model Comparison, Machine Learning Fundamentals, Kaggle Competition Experience, Performance Evaluation.**
Jana Saati	**Model training, fine-tuning, and result analysis**	**Deep learning, transfer learning, performance analysis**
Bayader Alfaifi	**Data preprocessing and image preparation**	**Data cleaning, data augmentation, dataset management**